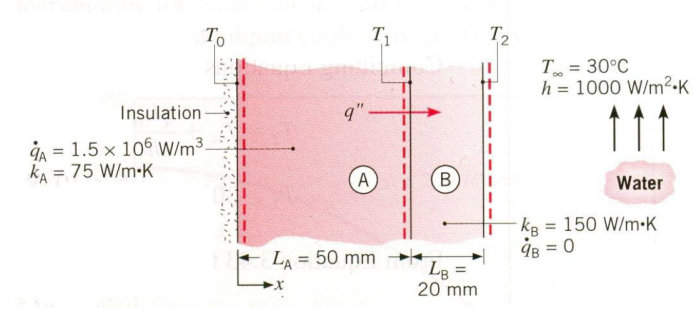


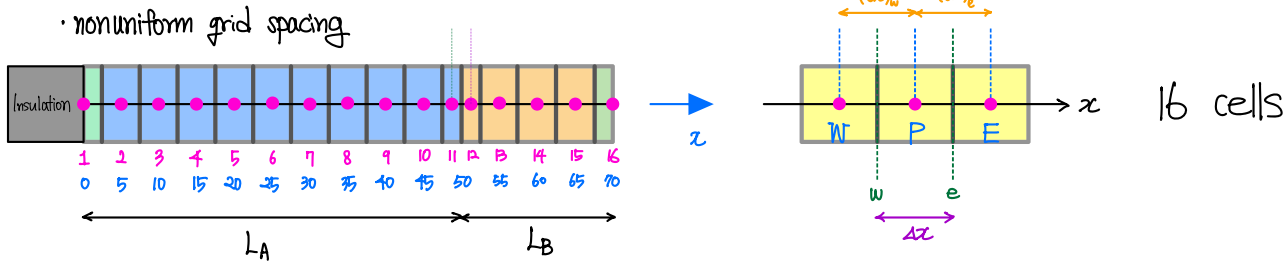
CFD HW #4 기계공학과 편혁준

1. A plane wall is a composite of two materials, A and B. The wall of material A has uniform heat generation $\dot{q} = 1.5 \times 10^6 \text{ W/m}^3$, $k_A = 75 \text{ W/m}\cdot\text{K}$, and thickness $L_A = 50 \text{ mm}$. The wall material B has no generation with $k_B = 150 \text{ W/m}\cdot\text{K}$ and thickness $L_B = 20 \text{ mm}$. The inner surface of material A is well insulated, while the outer surface of material B is cooled by a water stream with $T_\infty = 30^\circ\text{C}$ and $h = 1000 \text{ W/m}^2\cdot\text{K}$.



- (a) Solve numerically for the temperature distribution that exists in the composite under steady state conditions.
(b) Determine the temperature T_0 of the insulated surface and the temperature T_2 of the cooled surface.

* Step 1 : Discretize the domain



* Step 2 : Integrate the differential equation

$$\text{Differential equation} = \frac{d}{dx} \left(k \frac{dT}{dx} \right) + \dot{S} = 0$$

$$\int_w^e \left\{ \frac{d}{dx} \left(k \frac{dT}{dx} \right) + \dot{S} \right\} dx = \left(k \frac{dT}{dx} \right)_e - \left(k \frac{dT}{dx} \right)_w + \int_w^e \dot{S} dx = 0$$

* Step 3 : Select piecewise profile and formulate algebraic equation.

- piecewise linear profile

$$\left(k \frac{dT}{dx} \right)_e - \left(k \frac{dT}{dx} \right)_w + \int_w^e \dot{S} dx = k_e \frac{T_E - T_P}{\delta x_e} - k_w \frac{T_P - T_W}{\delta x_w} + \bar{S} \cdot \Delta x = 0$$

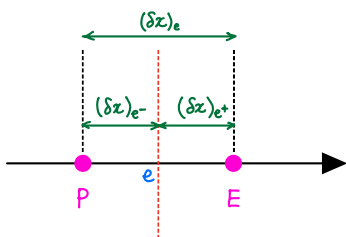
- Source term linearization

$$\bar{S} = S_c + \frac{S_p T_P}{b} = S_c \quad \left(S_c = \begin{cases} \dot{q} & (0 \leq x \leq 50) \\ 0 & (50 < x \leq 70) \end{cases} \right)$$

$$a_E = \frac{k_e}{(\delta x)_e}, \quad a_W = \frac{k_w}{(\delta x)_w}, \quad a_P = a_E + a_W, \quad b = S_c \Delta x \rightarrow a_P T_P = a_E T_E + a_W T_W + b : \text{Discretization equation}$$

* The interface conductivity

by electric circuit analogy,



$$g_e = \frac{T_P - T_E}{(\delta x)_e/k_P + (\delta x)_e^*/k_E} \rightarrow \frac{(\delta x)_e}{k_e} = \frac{(\delta x)_e}{k_P} + \frac{(\delta x)_e^*}{k_E}$$

$$k_e = \left(\frac{1-f_e}{k_P} + \frac{f_e}{k_E} \right)^{-1}$$

$$\left(f_e \equiv \frac{(\delta x)_e^*}{(\delta x)_e}, \quad 1-f_e \equiv \frac{(\delta x)_e}{(\delta x)_e} \right)$$

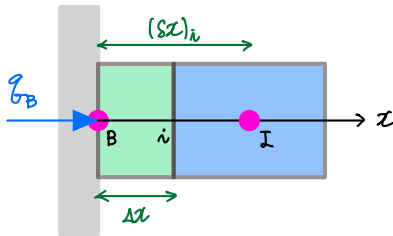
In the current discretized domain, the value of f on the surface "e" is as follows.

$$(\delta x)_e = 2.5 \text{ mm}, (\delta x)_{e^+} = 1.25 \text{ mm}$$

$$f_e = \frac{(\delta x)_{e^+}}{(\delta x)_e} = \frac{1}{2}$$

$$\rightarrow K_e = \left(\frac{1/2}{K_P} + \frac{1/2}{K_E} \right)^{-1} = \frac{2K_P K_E}{K_E + K_P}$$

• left boundary



Node #1 #2
 $\Delta x = 0.0025$
 $(\delta x)_i = 0.005$

$$\frac{d}{dx} \left(K \frac{dT}{dx} \right) + S' = 0$$

$$K_i \left(\frac{dT}{dx} \right)_i - K_B \left(\frac{dT}{dx} \right)_B + (S_c + S_p T_B) \Delta x = 0$$

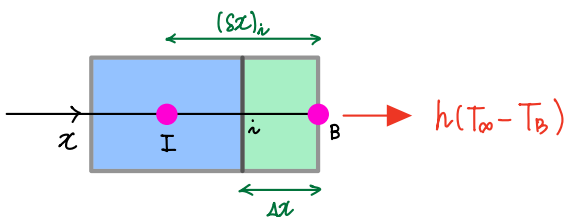
$$\frac{K_i (T_i - T_B)}{(\delta x)_i} + q_B + (S_c + S_p T_B) \Delta x = 0$$

$$\left(\frac{K_i}{(\delta x)_i} - S_p \Delta x \right) T_B = \frac{K_i}{(\delta x)_i} T_i + S_c \Delta x + q_B, \quad K_i = \frac{2 \times K_A \times K_A}{K_A + K_A} = K_A$$

$$\left(\begin{array}{l} q_B = 0 \\ S_p = 0 \\ S_c = \dot{q}_A \end{array} \right) \rightarrow \frac{K_A}{(0.005)} T_i = \frac{K_A}{(0.005)} T_i + (0.0025) \dot{q}_A = 0$$

$\downarrow$ $\downarrow$ $\downarrow$
 a_1 a_2 b_1

• right boundary



Node #15 #16

$$\frac{d}{dx} \left(K \frac{dT}{dx} \right) + S' = 0$$

$$h(T_\infty - T_B) - K_i \left(\frac{dT}{dx} \right)_i + (S_c + S_p T_B) \Delta x = 0$$

$$h(T_\infty - T_B) - \frac{K_i (T_B - T_i)}{(\delta x)_i} + (S_c + S_p T_B) \Delta x = 0$$

$$\left(\frac{K_i}{(\delta x)_i} + h - S_p \Delta x \right) T_B = \frac{K_i}{(\delta x)_i} T_i + S_c \Delta x + h T_\infty, \quad K_i = K_B$$

$$S' = 0 \rightarrow -h(T_{16} - T_\infty) - \frac{K_B (T_{16} - T_{15})}{(0.005)} = 0$$

$(S_p, S_c = 0)$

$$\left(\frac{K_B}{(0.005)} + h \right) T_{16} = \frac{K_B}{(0.005)} T_{15} + h T_\infty$$

$\downarrow$ $\downarrow$ $\downarrow$
 a_{16} a_{15} b_{16}

- internal nodes

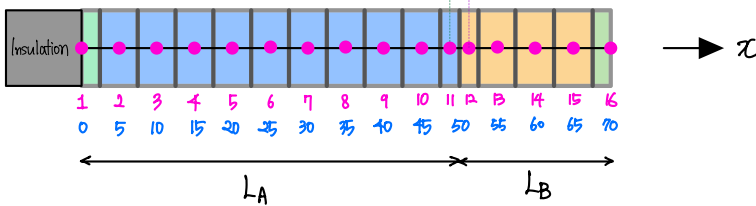
$$\left(k \frac{dT}{dx} \right)_e - \left(k \frac{dT}{dx} \right)_w + \int_w^e \dot{S} dx = k_e \frac{T_E - T_P}{\delta x_e} - k_w \frac{T_P - T_W}{\delta x_w} + \bar{S} \cdot \Delta x = 0$$

$$\bar{S} = S_c + \underbrace{S_p T_P}_{\rightarrow 0} = S_c \quad \left(S_c = \begin{cases} \dot{q} & (0 \leq x \leq 50) \\ 0 & (50 < x \leq 100) \end{cases} \right)$$

$$a_E = \frac{k_e}{(\delta x)_e}, \quad a_W = \frac{k_w}{(\delta x)_w}, \quad a_P = a_E + a_W, \quad b = S_c \Delta x \rightarrow a_P T_P = a_E T_E + a_W T_W + b$$

$$k_e = \left(\frac{1-f_e}{k_P} + \frac{f_e}{k_E} \right)^{-1} \quad \left(f_e \equiv \frac{(\delta x)_e^+}{(\delta x)_e}, \quad 1-f_e \equiv \frac{(\delta x)_e^-}{(\delta x)_e} \right)$$

$$f_e = \frac{(\delta x)_e^+}{(\delta x)_e} = \frac{1}{2} \rightarrow k_e = \left(\frac{1/2}{k_P} + \frac{1/2}{k_E} \right)^{-1} = \frac{2k_P k_E}{k_E + k_P}$$



i) Node 2 ~ 9

$$\left(\frac{k_A}{0.005} + \frac{k_A}{0.005} \right) T_i = \frac{k_A}{0.005} T_{i+1} + \frac{k_A}{0.005} T_{i-1} + \dot{q}_A \cdot (0.005)$$

ii) Node 10

$$\left(\frac{k_A}{0.00375} + \frac{k_A}{0.005} \right) T_{10} = \frac{k_A}{0.00375} T_{i+1} + \frac{k_A}{0.005} T_{i-1} + \dot{q}_A \cdot (0.005)$$

iii) Node 11

$$\left(\frac{\frac{2k_A k_B}{k_B + k_A}}{0.0025} + \frac{k_A}{0.00375} \right) T_{11} = \frac{\frac{2k_A k_B}{k_B + k_A}}{0.0025} T_{12} + \frac{k_A}{0.00375} T_{10} + \dot{q}_A \cdot (0.0025)$$

iv) Node 12

$$\left(\frac{k_B}{0.00375} + \frac{\frac{2k_A k_B}{k_B + k_A}}{0.0025} \right) T_{12} = \frac{k_B}{0.00375} T_{13} + \frac{\frac{2k_A k_B}{k_B + k_A}}{0.0025} T_{11} + 0$$

v) Node 13

$$\left(\frac{k_B}{0.005} + \frac{k_B}{0.00375} \right) T_{13} = \frac{k_B}{0.005} T_{14} + \frac{k_B}{0.00375} T_{12} + 0$$

vi) Node 14 ~ 15

$$\left(\frac{k_B}{0.005} + \frac{k_B}{0.005} \right) T_i = \frac{k_B}{0.005} T_{i+1} + \frac{k_B}{0.005} T_{i-1} + 0$$

- The four basic rules

1) Rule #1 : Consistency at control volume surfaces

→ It is satisfied since I used harmonic mean conductivity formulation over the entire interfaces.

2) Rule #2 : Positive coefficients

→ It is satisfied since a_E, a_W , and a_p are always positive.

a_p is always positive because $\bar{S} = S_c$ ($S_p = 0$)

3) Rule #3 : Negative - slope linearization

→ It is satisfied since the source term has only constant value.

Since $\bar{S}$ is equal to S_c , S_p is always zero. ($\bar{S} = S_c + S_p T_p$)

4) Sum of the neighbor coefficients ($a_p = \sum a_{nb}$)

→ It is satisfied since the differential equation we are dealing with is as follows.

$$\frac{d}{dx} \left(k \frac{dT}{dx} \right) + \bar{S} = 0 \quad \left(\bar{S} = \begin{cases} \dot{q}_A & (0 \leq x \leq 50) \\ 0 & (50 < x \leq 100) \end{cases} \right)$$

We only have derivative term and source term with constant value.

So the differential equation continues to remain satisfied after a constant is added to the dependant variable.

Thus, $a_p = \sum a_{nb}$ is always satisfied.

- Scarborough criterion

$$a_E = \frac{k_e}{(\delta x)_e}, \quad a_W = \frac{k_w}{(\delta x)_w}, \quad a_p = a_E + a_W, \quad b = S_c \Delta x \quad \longrightarrow \quad a_p T_p = a_E T_E + a_W T_W + b$$

$\hookrightarrow a_p = \sum a_{nb}$

$$\frac{\sum |a_{nb}|}{|a_p|} \leq 1 \quad \text{is satisfied. So, it is sufficient for convergence.}$$

* Step 4. Solve the discretized equation.

$$a_p T_p = a_E T_E + a_W T_W + b \quad \longrightarrow \quad \text{TDMA Algorithm}$$

The above equations can be written as

$$a_i T_i = b_i T_{i+1} + c_i T_{i-1} + d_i \quad (i = 1, 2, \dots, N)$$

To account for the boundary conditions,

$$i = 1 \longrightarrow c_1 = 0, \quad i = N \longrightarrow b_N = 0$$

In TDMA method, P_i and Q_i are introduced to explain forward-substitution. ($T_i = P_i T_{i+1} + Q_i$)

$$P_i = \frac{b_i}{a_i - c_i P_{i-1}}, \quad Q_i = \frac{d_i + c_i Q_{i-1}}{a_i - c_i P_{i-1}} \quad (i = 2, 3, 4, \dots, N)$$

$$i = 1 \longrightarrow P_1 = \frac{b_1}{a_1}, \quad Q_1 = \frac{d_1}{a_1} \quad (\because c_1 = 0) \quad P_N = 0 \quad (\because b_N = 0) \longrightarrow T_N = Q_N$$

After finishing forward substitution, back-substitution is now ready.

$$T_i = P_i T_{i+1} + Q_i \quad (i = N-1, N-2, \dots, 1)$$

When back-substitution is over, we can obtain the solution.

```

1  %% Problem 1: A plane wall composed of two materials
2  clear; clc; close all;
3
4  %% Parameters
5  kA      = 75;           % W/(m·K)
6  kB      = 150;          % W/(m·K)
7  qdot_A  = 1.5e6;        % W/m^3 (heat generation at region A)
8  LA      = 0.05;         % m
9  LB      = 0.02;         % m
10 h       = 1000;         % W/(m^2·K)
11 Tinf    = 303.15;       % K (30°C)
12
13 %% Discretization of the domain
14 % Region A (0 to 50 mm): 11 cells, Region B (50 to 70 mm): 5 cells
15 nA = 11;
16 nB = 5;
17 Num_cells = nA + nB; % total: 16 cells
18
19 % Cell width (m)
20 w = zeros(Num_cells,1);
21 w(1) = 0.0025;
22 w(2:10) = 0.005;
23 w(11) = 0.0025;
24 w(12) = 0.0025;
25 w(13:15) = 0.005;
26 w(16) = 0.0025;
27
28 %% Location of the entire interfaces
29 Interfaces = [0; cumsum(w)]; % 17 elements: including both boundaries
30
31 %% Location of each node
32 % For boundary cells, node is at the boundary; for interior cells, at cell center.
33 node = zeros(Num_cells,1);
34 for i = 1:Num_cells
35     if i==1
36         node(i) = Interfaces(1); % left boundary at 0 m
37     elseif i==Num_cells
38         node(i) = Interfaces(end); % right boundary at 0.07 m
39     else
40         node(i) = (Interfaces(i) + Interfaces(i+1))/2;
41     end
42 end
43 % Example: node = [0; 0.005; 0.01; ... ; 0.07]
44
45 %% Allocation of property for each cell
46 k = zeros(Num_cells,1); % thermal conductivity
47 dx = w; % cell width = w
48 heat_gen = zeros(Num_cells,1); % heat generation (q_dot)
49 for i = 1:Num_cells
50     if i <= nA % Region A
51         k(i) = kA;
52         heat_gen(i) = qdot_A;
53     else % Region B
54         k(i) = kB;
55         heat_gen(i) = 0;
56     end
57 end
58
59 %% Distance between nodes (used for conduction coefficient)
60 halfWidth = dx/2; % half-width for each cell
61 Num_faces = Num_cells - 1;
62 d_nodes = zeros(Num_faces,1); % distance between adjacent cell centers
63 for i = 1:Num_faces
64     if i == 1
65         % Left boundary
66         d_nodes(i) = dx(1) + halfWidth(i+1);
67     elseif i == Num_faces
68         % Right boundary
69         d_nodes(i) = halfWidth(i) + dx(i+1);
70     else
71         d_nodes(i) = halfWidth(i) + halfWidth(i+1);
72     end
73 end
74

```

```

76 %% Compute conduction coefficients, C = k_face / d_nodes
77 % For each face, use harmonic mean for conductivity.
78 C = zeros(Num_faces,1);
79 for i = 1:Num_faces
80     if i == 11 % interface between cell 11 (Region A) and cell 12 (Region B)
81         k_face = 2*kA*kB/(kA + kB);
82     else
83         k_face = 2*k(i)*k(i+1)/(k(i) + k(i+1));
84     end
85     C(i) = k_face / d_nodes(i);
86 end
87

```

```

88 %% Assemble FVM equations using aP, aW, aE vectors
89
90 % For interior cells (i = 2 to 15):
91 % -C(i-1)*T(i-1) + [C(i-1)+C(i)]*T(i) - C(i)*T(i+1) = heat_gen(i)*dx(i)
92 % Left boundary (cell 1, insulated):
93 % C(1)*(T(1) - T(2)) = heat_gen(1)*dx(1)
94 % Right boundary (cell 16, convective):
95 % -C(15)*(T(15)-T(16)) + h*(T(16)-Tinf) = 0
96 %
97 aP = zeros(Num_cells,1);
98 aW = zeros(Num_cells,1);
99 aE = zeros(Num_cells,1);
100 b_vec = zeros(Num_cells,1);
101
102 % Left boundary (insulated, i=1)
103 aP(1) = C(1);
104 aE(1) = -C(1);
105 b_vec(1) = heat_gen(1)*dx(1);
106
107 % Interior cells (i = 2 to Ncells-1 = 15)
108 for i = 2:Num_cells-1
109     aW(i) = -C(i-1);
110     aP(i) = C(i-1) + C(i);
111     aE(i) = -C(i);
112     b_vec(i) = heat_gen(i)*dx(i);
113 end
114
115 % Right boundary (convection, i=16)
116 aW(Num_cells) = -C(Num_faces);
117 aP(Num_cells) = C(Num_faces) + h;
118 aE(Num_cells) = 0;
119 b_vec(Num_cells) = heat_gen(Num_cells)*dx(Num_cells) + h*Tinf;
120

```

```

121 %% Solve the linear system using TDMA
122 % aW (sub-diagonal), aP (diagonal), aE (super-diagonal)
123 T_tdma = TDMA_solver(aP, aW, aE, b_vec);
124

```

```

125 %% TDMA Solver
126 function T = TDMA_solver(aP, aW, aE, b)
127     n = length(b); % 16
128     T = zeros(n,1);
129     P = zeros(n,1);
130     Q = zeros(n,1);
131
132     % first row manipulation (i=1)
133     P(1) = aE(1) / aP(1);
134     Q(1) = b(1) / aP(1);
135
136     % Front Substitution
137     for i = 2:n
138         denom = aP(i) - aW(i) * P(i-1);
139         if i < n
140             P(i) = aE(i) / denom;
141         end
142         Q(i) = (b(i) - aW(i) * Q(i-1)) / denom;
143     end
144
145     % Back Substitution
146     T(n) = Q(n);
147     for i = n-1:-1:1
148         T(i) = Q(i) - P(i) * T(i+1);
149     end
150 end
151

```

(a) The temperature distribution under steady state conditions

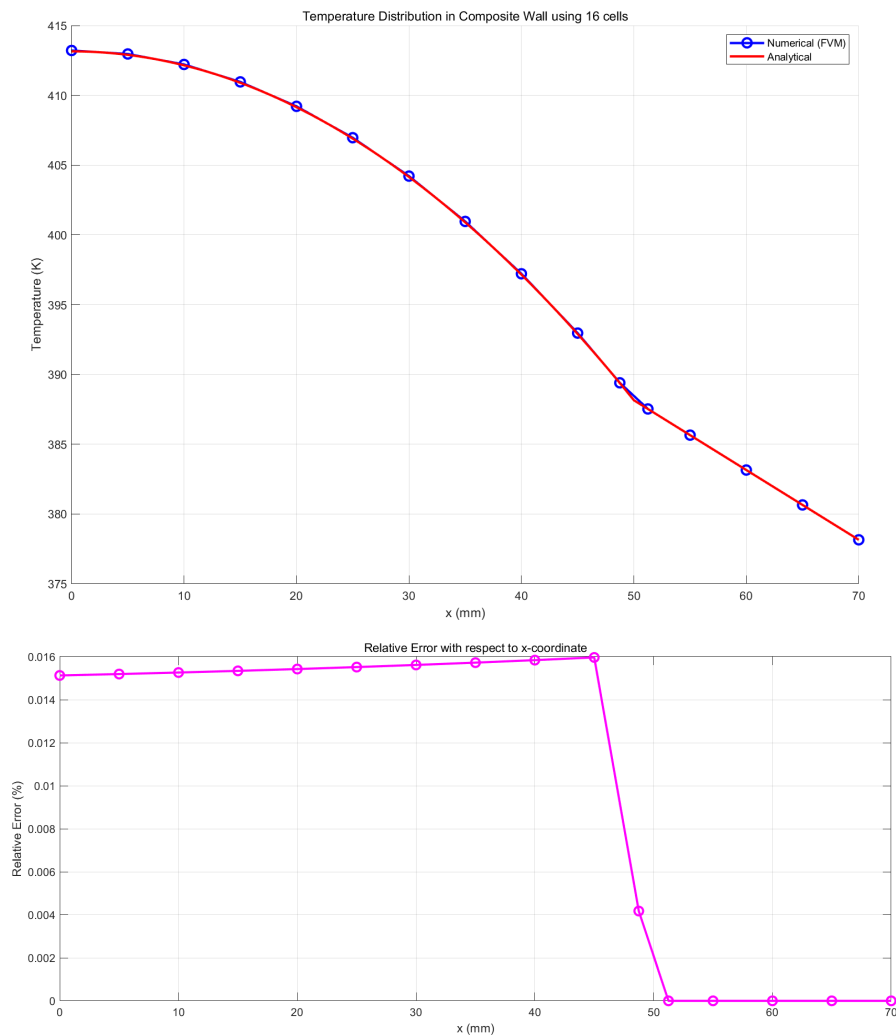
Node	x (m)	Temperature (K)	Analytical (K)	Abs Error (K)	Rel Error (%)
1	0.000000	413.21	413.15	0.06	0.02
2	0.005000	412.96	412.90	0.06	0.02
3	0.010000	412.21	412.15	0.06	0.02
4	0.015000	410.96	410.90	0.06	0.02
5	0.020000	409.21	409.15	0.06	0.02
6	0.025000	406.96	406.90	0.06	0.02
7	0.030000	404.21	404.15	0.06	0.02
8	0.035000	400.96	400.90	0.06	0.02
9	0.040000	397.21	397.15	0.06	0.02
10	0.045000	392.96	392.90	0.06	0.02
11	0.048750	389.40	389.38	0.02	0.00
12	0.051250	387.52	387.52	0.00	0.00
13	0.055000	385.65	385.65	0.00	0.00
14	0.060000	383.15	383.15	0.00	0.00
15	0.065000	380.65	380.65	0.00	0.00
16	0.070000	378.15	378.15	0.00	0.00

RMS Error = 0.0499 K

$$T_0 = 413.21 \text{ K}$$

$$T_1 = \frac{389.4 + 387.52}{2} = 388.46 \text{ K}$$

$$T_2 = 378.15 \text{ K}$$



The numerical solution obtained using 16 cells showed an RMS error of 0.049 when compared to the analytical solution.

This suggests that even with a small number of cells, the results nearly match the exact solution.

The temperature T_1 at the insulated wall was calculated to be 413.21 K, and the temperature T_{16} at the convective boundary was calculated to be 378.15 K. In region A, where the source term is present, the temperature decreases gradually; however, in region B, where the source term is absent, the temperature decreases linearly.

Additionally, when the source term is present, a relative error of approximately 0.02 % occurred, whereas the error in the region without the source term was calculated to be zero.

(b) The temperature T_0 of the insulated surface and the temperature T_2 of the cooled surface.

Total length of refined grid: 0.0700 m

Node	x (m)	Temperature (K)	Analytical (K)	Abs Error (K)	Rel Error (%)
1	0.000000	413.15	413.15	0.00	0.00
2	0.001250	413.14	413.13	0.00	0.00
3	0.002500	413.09	413.09	0.00	0.00
4	0.003750	413.01	413.01	0.00	0.00
5	0.005000	412.90	412.90	0.00	0.00
6	0.006250	412.76	412.76	0.00	0.00
7	0.007500	412.59	412.59	0.00	0.00
8	0.008750	412.39	412.38	0.00	0.00
9	0.010000	412.15	412.15	0.00	0.00
10	0.011250	411.89	411.88	0.00	0.00
11	0.012500	411.59	411.59	0.00	0.00
12	0.013750	411.26	411.26	0.00	0.00
13	0.015000	410.90	410.90	0.00	0.00
14	0.016250	410.51	410.51	0.00	0.00
15	0.017500	410.09	410.09	0.00	0.00
16	0.018750	409.64	409.63	0.00	0.00
17	0.020000	409.15	409.15	0.00	0.00
18	0.021250	408.64	408.63	0.00	0.00
19	0.022500	408.09	408.09	0.00	0.00
20	0.023750	407.51	407.51	0.00	0.00
21	0.025000	406.90	406.90	0.00	0.00
22	0.026250	406.26	406.26	0.00	0.00
23	0.027500	405.59	405.59	0.00	0.00
24	0.028750	404.89	404.88	0.00	0.00
25	0.030000	404.15	404.15	0.00	0.00
26	0.031250	403.39	403.38	0.00	0.00
27	0.032500	402.59	402.59	0.00	0.00
28	0.033750	401.76	401.76	0.00	0.00
29	0.035000	400.90	400.90	0.00	0.00
30	0.036250	400.01	400.01	0.00	0.00
31	0.037500	399.09	399.09	0.00	0.00
32	0.038750	398.14	398.13	0.00	0.00
33	0.040000	397.15	397.15	0.00	0.00
34	0.041250	396.14	396.13	0.00	0.00
35	0.042500	395.09	395.09	0.00	0.00
36	0.043750	394.01	394.01	0.00	0.00
37	0.045000	392.90	392.90	0.00	0.00
38	0.046250	391.76	391.76	0.00	0.00
39	0.047500	390.59	390.59	0.00	0.00
40	0.048750	389.39	389.38	0.00	0.00
41	0.049688	388.46	388.46	0.00	0.00
42	0.050313	387.99	387.99	0.00	0.00
43	0.051250	387.52	387.52	0.00	0.00
44	0.052500	386.90	386.90	0.00	0.00
45	0.053750	386.27	386.27	0.00	0.00
46	0.055000	385.65	385.65	0.00	0.00
47	0.056250	385.02	385.02	0.00	0.00
48	0.057500	384.40	384.40	0.00	0.00
49	0.058750	383.77	383.77	0.00	0.00
50	0.060000	383.15	383.15	0.00	0.00
51	0.061250	382.52	382.52	0.00	0.00
52	0.062500	381.90	381.90	0.00	0.00
53	0.063750	381.27	381.27	0.00	0.00
54	0.065000	380.65	380.65	0.00	0.00
55	0.066250	380.02	380.02	0.00	0.00
56	0.067500	379.40	379.40	0.00	0.00
57	0.068750	378.77	378.77	0.00	0.00
58	0.070000	378.15	378.15	0.00	0.00

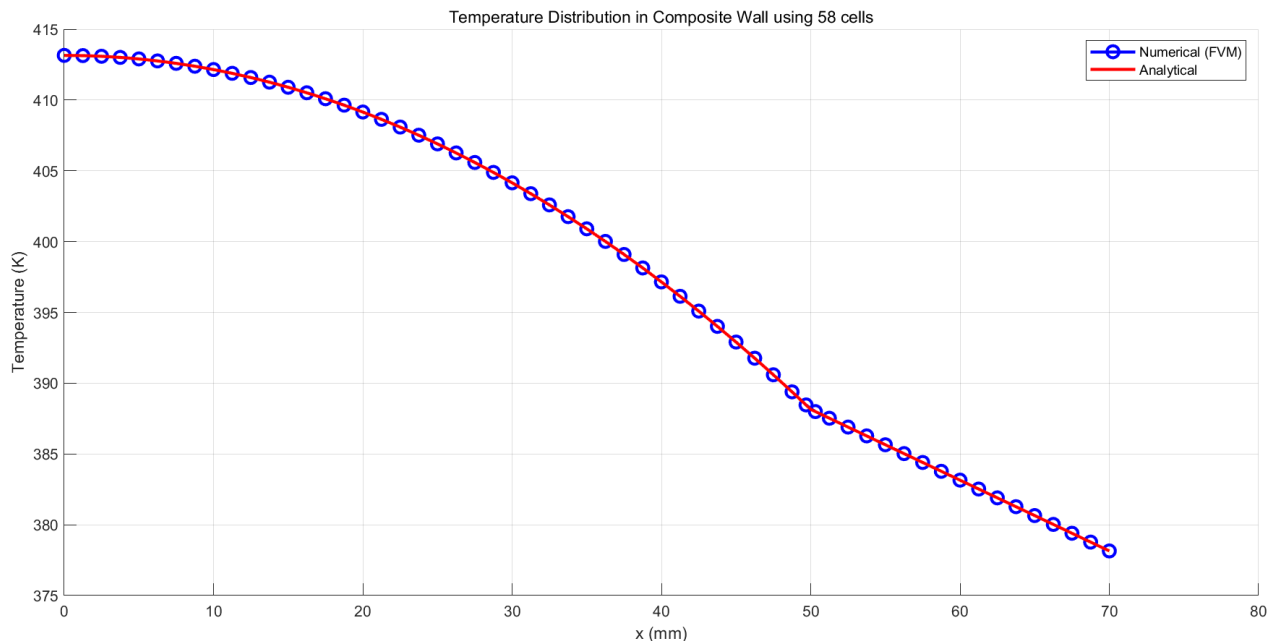
RMS Error = 0.0036 K

When the number of cells was increased from 16 to 58 and the numerical solution was obtained again, the result is as follows. (←)

The temperature of the insulated surface is 413.15 K (= 140°C)

and the temperature of the cooled surface is 378.15 K (= 105°C).

I obtained the same result as the analytical solution at T_0 and T_2 .



C.f.) Analytical solution

1D Steadystate energy equation

$$A: \frac{d^2 T}{dx^2} = -\frac{\dot{q}_A}{k_A} \quad (0 \leq x \leq L_A)$$

$$\frac{dT}{dx} = -\frac{\dot{q}_A}{k_A} x + C_1 \quad (C_1: \text{constant})$$

$$T(x) = -\frac{\dot{q}_A}{2k_A} x^2 + C_1 x + C_2$$

$$B: \frac{d^2 T}{dx^2} = 0 \quad (L_A < x \leq L_B)$$

$$T(x) = C_3 x + C_4$$

$$\text{B.C.s } ① \left. \frac{dT}{dx} \right|_{x=0} = 0 \rightarrow C_1 = 0$$

$$② -k_A \left. \frac{dT}{dx} \right|_{x=L_A^-} = -k_B \left. \frac{dT}{dx} \right|_{x=L_A^+} \rightarrow \dot{q}_A L_A = -k_B \cdot C_3 \rightarrow C_3 = -\frac{\dot{q}_A L_A}{k_B}$$

$$T_A(L_A) = T_B(L_A) \rightarrow -\frac{\dot{q}_A}{2k_A} L_A^2 + C_2 = C_3 \cdot L_A + C_4 \rightarrow C_2 = \frac{\dot{q}_A}{2k_A} L_A^2 - \frac{\dot{q}_A L_A^2}{k_B} + C_4$$

$$③ -k_B \left. \frac{dT}{dx} \right|_{x=L_A+L_B} = h(T_2 - T_\infty) \rightarrow -k_B \cdot C_3 = -k_B \cdot \left(-\frac{\dot{q}_A L_A}{k_B}\right) = h(T_2 - T_\infty)$$

$$T_2 = C_3 \cdot (L_A + L_B) + C_4$$

$$\rightarrow \dot{q}_A L_A = h \left\{ -\frac{\dot{q}_A L_A}{k_B} \cdot (L_A + L_B) + C_4 - T_\infty \right\}$$

$$C_4 = \frac{\dot{q}_A}{h} L_A + \frac{\dot{q}_A L_A}{k_B} \cdot (L_A + L_B) + T_\infty$$

$$C_3 = -\frac{\dot{q}_A L_A}{k_B} = -\frac{(1.5 \times 10^6 \text{ W/m}^3) \cdot (50 \times 10^{-3} \text{ m})}{150 \text{ W/m}\cdot\text{K}} = -500 \text{ K/m}$$

$$C_2 = \frac{\dot{q}_A}{2k_A} L_A^2 - \frac{\dot{q}_A L_A^2}{k_B} + C_4 = (1.5 \times 10^6 \text{ W/m}^3) \cdot (50 \times 10^{-3} \text{ m})^2 \left(\frac{1}{2 \cdot 175 \text{ W/m}\cdot\text{K}} - \frac{1}{150 \text{ W/m}\cdot\text{K}} \right) + C_4 \rightarrow C_2 = C_4$$

$$C_4 = \frac{\dot{q}_A}{h} L_A + \frac{\dot{q}_A L_A}{k_B} \cdot (L_A + L_B) + T_\infty = (1.5 \times 10^6 \text{ W/m}^3) \cdot (50 \times 10^{-3} \text{ m}) \left\{ \frac{1}{(1000 \text{ W/m}^2\cdot\text{K})} + \frac{(10 \times 10^{-3} \text{ m})}{(150 \text{ W/m}\cdot\text{K})} \right\} + 303.15 \text{ K}$$

$$= 413.15 \text{ K}$$

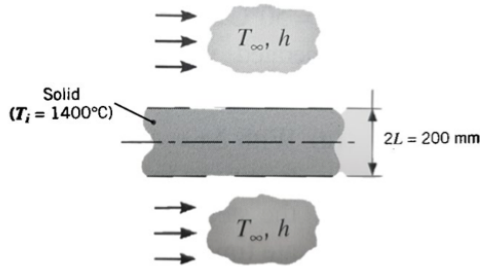
$$T_A(x) = -\frac{\dot{q}_A}{2k_A} x^2 + C_1 x + C_2 = -\frac{(1.5 \times 10^6 \text{ W/m}^3)}{2 \cdot (175 \text{ W/m}\cdot\text{K})} x^2 + 413.15 \text{ K} = -10000 x^2 + 413.15 \text{ (K)}$$

$$T_B(x) = C_3 x + C_4 = -500x + 413.15 \text{ (K)}$$

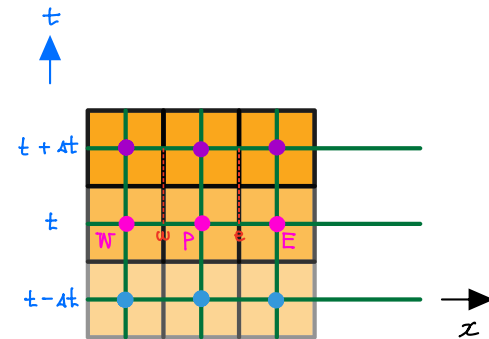
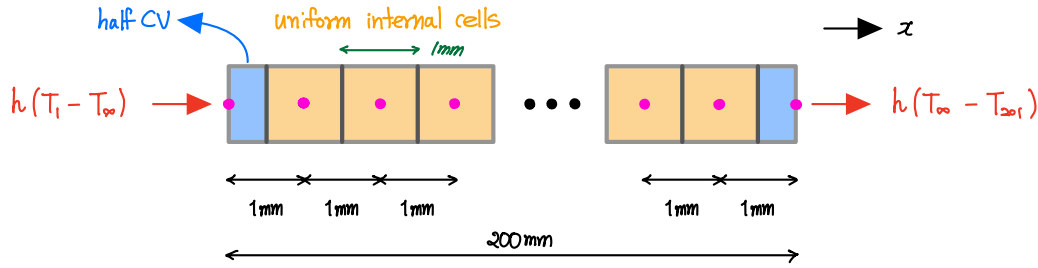
$$T(x) = \begin{cases} -10000x^2 + 413.15 \text{ (K)} & (0 \leq x \leq 0.05 \text{ m}) \\ -500x + 413.15 \text{ (K)} & (0.05 \text{ m} < x < 0.07 \text{ m}) \end{cases}$$

$$\rightarrow T_0 = 413.15 \text{ K} = 140^\circ\text{C}, \quad T_2 = -500(0.07) + 413.15 = 378.15 \text{ K} = 105^\circ\text{C}$$

2. Consider a 200-mm-thick solid slab of steel ($\rho = 7800 \text{ kg/m}^3$, $c = 700 \text{ J/kg} \cdot \text{K}$, $k = 30 \text{ W/m} \cdot \text{K}$), initially at a uniform temperature of $T_i = 1400^\circ\text{C}$. The slab is cooled at its top and bottom surfaces by water jets ($T_\infty = 50^\circ\text{C}$), which maintain an approximately uniform convection coefficient $h = 5000 \text{ W/m}^2 \cdot \text{K}$ at both surfaces.
- (a) Using the fully-implicit numerical method with a space increment of $\Delta x = 1 \text{ mm}$, determine the time required to cool the surface of the slab to 300°C .
- (b) What is the corresponding temperature at the midplane of the slab?



* Step 1. Discretize the domain.



According to the problem, to achieve a Δx of 1 mm, the domain was discretized using two half control volumes (each of 0.5 mm in size) and 199 full cells (each of 1 mm in size).

* Step 2. Integrate the governing differential equation over the CV and over the time interval.

$$\text{Differential equation: } \rho c \frac{\partial T}{\partial t} = \frac{\partial}{\partial x} \left(k \frac{\partial T}{\partial x} \right) \rightarrow \text{parabolic PDE}$$

$$\xrightarrow{\text{integration}} \rho c \int_w^e \int_t^{t+\Delta t} \frac{\partial T}{\partial t} dt dx = \int_w^e \int_t^{t+\Delta t} \frac{\partial}{\partial x} \left(k \frac{\partial T}{\partial x} \right) dt dx \quad (\rho, c : \text{constant})$$

* Step 3. Select proper piecewise profile and formulate algebraic equation.

Let's assume T_p prevails throughout the CV and employ piecewise linear profile for space variation.

$$\text{LHS: } \rho c \int_w^e (T^{t+\Delta t} - T^t) dx = \rho c (T_p^t - T_p^o) \Delta x$$

$$\text{RHS: } \int_t^{t+\Delta t} \left[k_e \frac{T_E - T_P}{(\Delta x)_e} - k_w \frac{T_P - T_W}{(\Delta x)_w} \right] dt$$

According to the problem, I have to use fully-implicit method.

By assuming exponential variation with respect to time, we can get discretized equation as follows.

$$\int_t^{t+\Delta t} T_P dt = T_P' \cdot \Delta t = T_P \cdot \Delta t \quad (\text{dropping the superscript 1})$$

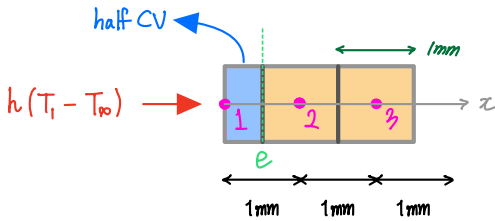
$$\longrightarrow \rho C (T_P - T_P^0) \Delta x = \left[k_e \frac{T_E - T_P}{(\Delta x)_e} - k_w \frac{T_P - T_W}{(\Delta x)_w} \right] \cdot \Delta t$$

$$a_E = \frac{k_e}{(\Delta x)_e}, \quad a_W = \frac{k_w}{(\Delta x)_w}, \quad a_P^0 = \frac{\rho C \Delta x}{\Delta t}, \quad a_P = a_E + a_W + a_P^0 \quad (a_E, a_W, a_P > 0)$$

$$\longrightarrow a_P T_P = a_E T_E + a_W T_W + a_P^0 T_P^0 : \text{discretized equation}$$

$$k_e = \frac{2 \cdot k \cdot K}{k + K} = k \quad (\text{harmonic mean})$$

i) Left boundary

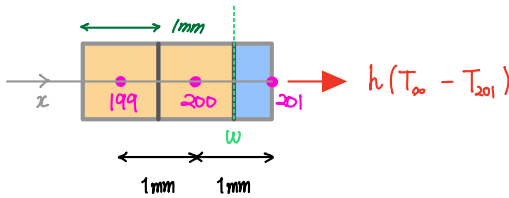


$$\rho C (T_1 - T_1^0) \cdot (0.001) = \left[\frac{k_e (T_2 - T_1)}{(0.001)} - h (T_1 - T_\infty) \right] \cdot \Delta t$$

$$\frac{k}{(0.001)} T_2 = \left\{ \frac{\rho C}{\Delta t} \cdot (0.001) + \frac{k}{(0.001)} + h \right\} T_1 - h T_\infty - \frac{\rho C T_1^0 \cdot (0.001)}{\Delta t}$$

$$k_w = \frac{2 \cdot k \cdot K}{k + K} = k \quad (\text{harmonic mean})$$

ii) Right boundary

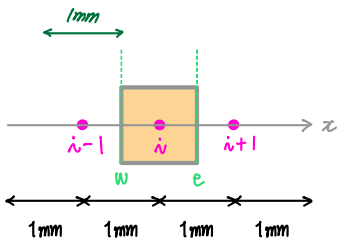


$$\rho C (T_{201} - T_{201}^0) \cdot (0.001) = \left[h (T_\infty - T_{201}) - \frac{k_w (T_{201} - T_{200})}{(0.001)} \right] \cdot \Delta t$$

$$\frac{\rho C (T_{201} - T_{201}^0) \cdot (0.001)}{\Delta t} = h (T_\infty - T_{201}) - \frac{k_w (T_{201} - T_{200})}{(0.001)}$$

$$\left(\frac{\rho C \cdot (0.001)}{\Delta t} + h + \frac{k}{(0.001)} \right) T_{201} = \frac{k}{(0.001)} T_{200} + h T_\infty + \frac{\rho C \cdot T_{201}^0 \cdot (0.001)}{\Delta t}$$

iii) Internal nodes



$$\rho C (T_i - T_i^0) (0.001) = \left[k_e \frac{T_{i+1} - T_i}{(0.001)} - k_w \frac{T_i - T_{i-1}}{(0.001)} \right] \cdot \Delta t$$

$$k_e = k_w = \frac{2 \cdot k \cdot K}{k + K} = k$$

$$\frac{k}{(0.001)} T_{i+1} = \left\{ \frac{k}{(0.001)} + \frac{k}{(0.001)} + \frac{\rho C \cdot (0.001)}{\Delta t} \right\} T_i$$

$$- \frac{k}{(0.001)} T_{i-1} - \frac{\rho C \cdot T_i^0 \cdot (0.001)}{\Delta t}$$

- The four basic rules

- 1) Rule #1 : Consistency at control volume surfaces

→ It is satisfied since it is composed of only one material and I'm going to use harmonic mean for k_e .
So, the conductivity over the entire interfaces are consistent.

- 2) Rule #2 : Positive coefficients

→ It is satisfied since a_E, a_W , and a_P are always positive. ($a_E, a_W, a_P > 0$)

- 3) Rule #3 : Negative - slope linearization

→ It is satisfied since the differential equation does not have a source term.

- 4) Sum of the neighbor coefficients ($a_P = \sum a_{nb}$)

→ It is satisfied since the differential equation we are dealing with is as follows.

$$\rho c \frac{dT}{dt} = \frac{d}{dz} \left(k \frac{dT}{dz} \right)$$

We only have derivative term and don't have any source term.

So the differential equation continues to remain satisfied after a constant is added to the dependant variable.

Thus, $a_P = \sum a_{nb}$ is always satisfied.

- Scarborough criterion

$$a_E = \frac{k_e}{(\Delta x)_e}, \quad a_W = \frac{k_w}{(\Delta x)_w}, \quad a_P^\circ = \frac{\rho c A \Delta x}{\Delta t}, \quad a_P = a_E + a_W + a_P^\circ \rightarrow a_P T_P = a_E T_E + a_W T_W + a_P^\circ T_P^\circ$$

$$\underline{a_P = \sum a_{nb} + a_P^\circ}$$

$$\frac{\sum |a_{nb}|}{|a_P|} \leq 1 \text{ is satisfied. So, it is sufficient for convergence.}$$

- * Step 4. Solve the discretized equation.

$$a_P T_P = a_E T_E + a_W T_W + b \rightarrow \text{TDMA Algorithm}$$

The above equations can be written as

$$a_i T_i = b_i T_{i+1} + c_i T_{i-1} + d_i \quad (i = 1, 2, \dots, N)$$

To account for the boundary conditions,

$$i = 1 \rightarrow c_1 = 0, \quad i = N \rightarrow b_N = 0$$

In TDMA method, P_i and Q_i are introduced to explain forward-substitution. ($T_i = P_i T_{i+1} + Q_i$)

$$P_i = \frac{b_i}{a_i - c_i P_{i-1}}, \quad Q_i = \frac{d_i + c_i Q_{i-1}}{a_i - c_i P_{i-1}} \quad (i = 2, 3, 4, \dots, N)$$

$$i = 1 \rightarrow P_1 = \frac{b_1}{a_1}, \quad Q_1 = \frac{d_1}{a_1} \quad (\because c_1 = 0) \quad P_N = 0 \quad (\because b_N = 0) \rightarrow T_N = Q_N$$

After finishing forward substitution, back-substitution is now ready.

$$T_i = P_i T_{i+1} + Q_i \quad (i = N-1, N-2, \dots, 1)$$

When back-substitution is over, we can obtain the solution.

< MATLAB >

```
% Problem 2: Slab Surface Cooling
clear; clc; close all;
```

```
%% Parameters
L      = 0.2;           % Slab thickness (m)
T_0    = 1400+273.15;   % Initial temperature (1400 °C)
T_inf  = 50+273.15;     % Ambient fluid temperature (50 °C)
h      = 5000;          % Convection heat transfer coefficient (W/m^2·K)
k      = 30;            % Thermal conductivity (W/m·K)
rho    = 7800;          % Density (kg/m^3)
cp     = 700;           % Specific heat (J/kg·K)
dx     = 0.001;         % Width of internal full cells (m)
N_cells_int = 199;      % Number of internal full cells
dt     = 0.01;          % Time step (s)
tmax   = 1000;          % Maximum simulation time (s)

% Criterion: surface temperature <= 300°C
T_surface_criterion = 300+273.15;
```

```
%% Grid Construction: half cell + full cells + half cell
Ncells = N_cells_int + 2; % Total number of cells (201)
w = zeros(Ncells,1);      % Cell widths

w(1) = dx / 2;            % Left boundary half cell
for i = 2:Ncells-1        % Internal full cells
    w(i) = dx;
end
w(Ncells) = dx / 2;       % Right boundary half cell

if abs(sum(w) - L) > 1e-12
    error('Total cell width does not equal to L (sum(w) = %g)', sum(w));
end

% Node positions (for reference)
b_faces = [0; cumsum(w)]; % Cell face coordinates
x_nodes = zeros(Ncells,1);
for i = 1:Ncells
    if i == 1
        x_nodes(i) = b_faces(1); % x = 0
    elseif i == Ncells
        x_nodes(i) = b_faces(end); % x = L
    else
        x_nodes(i) = 0.5*(b_faces(i) + b_faces(i+1)); % Cell center
    end
end
```

```
%% Time Integration + etc
% Initial Temperature Vector
T = T_0 * ones(Ncells,1); % 1400

% Coefficient
rcw = rho * cp * w; % (J/K)

% Conductive coefficients at cell faces
Nfaces = Ncells - 1;
C = zeros(Nfaces,1);

for i = 1:Nfaces
    dist = 0.5*w(i) + 0.5*w(i+1); % Distance between cell centers
    C(i) = k / dist;
end

time = 0;
step = 0;

% Variables for recording results
time_history = [];
T_history = []; % Stores temperature at all nodes

% Variables to store cooling criterion information
t_end = 0;
T_mid = 0; % Temperature at the midplane

mid_idx = round(Ncells/2); % Index of the midplane node
```

<pre> %% Fully Implicit Method while time < tmax time = time + dt; step = step + 1; % Coefficients for A * T^(1) = b aP = zeros(Ncells,1); aW = zeros(Ncells,1); aE = zeros(Ncells,1); b_vec = zeros(Ncells,1); %% Left Boundary (i=1): Convection + Conduction (neighbor: i=2) % rcw(1)*(T1^(1)-T1^(0)) / dt = C(1)*(T2^(1)-T1^(1)) - h*(T1^(1)-T_inf) aP(1) = rcw(1)/dt + C(1) + h; aE(1) = -C(1); b_vec(1) = rcw(1)/dt * T(1) + h*T_inf; %% Internal Cells (i=2 to Ncells-1) for i = 2:Ncells-1 % rcw(i)*(T_i^(1)-T_i^(0)) / dt = C(i-1)*(T_(i-1)^(1)-T_i^(1)) + C(i)*(T_(i+1)^(1)-T_i^(1)) aP(i) = rcw(i)/dt + C(i-1) + C(i); aW(i) = -C(i-1); aE(i) = -C(i); b_vec(i) = rcw(i)/dt * T(i); end %% Right Boundary (i=Ncells): Convection + Conduction (neighbor: i=Ncells-1) % rcw(Ncells)*(T_Ncells^(1)-T_Ncells^(0))/dt = C(Ncells-1)*(T_(Ncells-1)^(1)-T_Ncells^(1)) - h*(T_Ncells^(1)-T_inf) i = Ncells; aP(i) = rcw(i)/dt + C(i-1) + h; aW(i) = -C(i-1); b_vec(i) = rcw(i)/dt * T(i) + h*T_inf; %% Solve the linear system using TDMA T_new = TDMA_solver(aP, aW, aE, b_vec); T = T_new; % Record results time_history(end+1,1) = time; T_history(end+1,:) = T'; % Check if surface temperature meets the criterion if T(1) <= T_surface_criterion T(end) <= T_surface_criterion t_end = time; T_mid = T(mid_idx); break; end end end </pre>	
<pre> %% Convert Kelvin to Celsius for plotting and output T_C = T - 273.15; T_history_C = T_history - 273.15; % Convert all recorded temperatures fprintf('Required time to cool the surface to 300°C = %.3f s\n', t_end); fprintf('Temperature at the midplane = %.2f °C\n', T_mid - 273.15); </pre>	

* Results

```

Required time to cool the surface to 300°C = 55.220 s
Temperature at the midplane = 1399.93 °C
fx >>

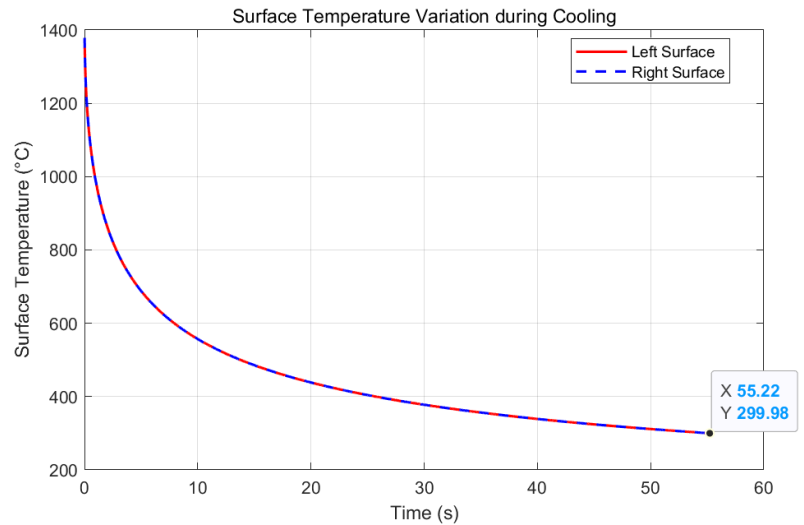
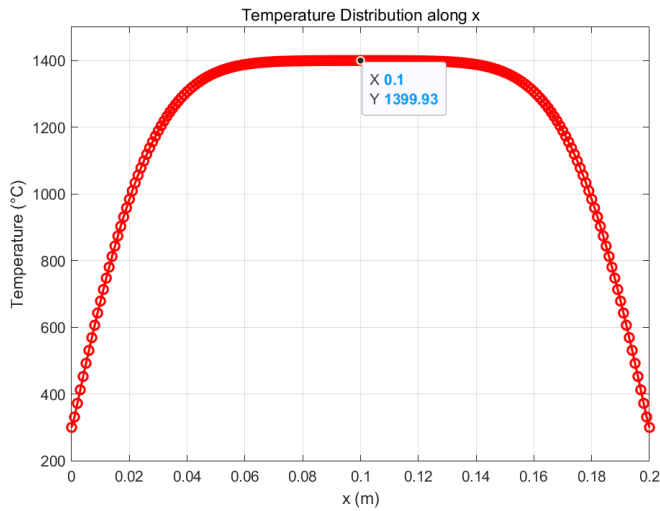
```

(a) It will take 55.2 seconds to cool the surface to 300 °C.

(b) The corresponding temperature at the midplane will be 1399.93 °C

* Discussion

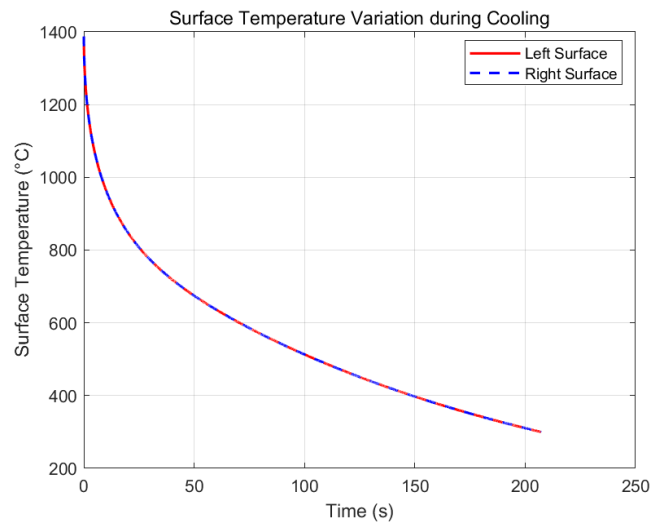
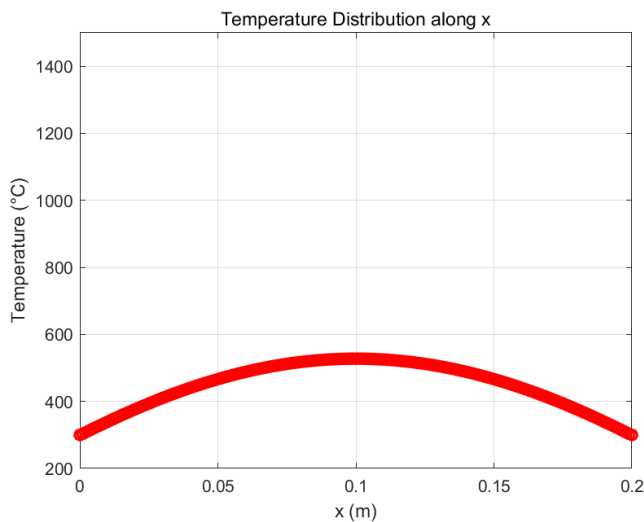
i) $K = 30 \text{ (W/m}\cdot\text{K)}$



Required time to cool the surface to 300°C = 55.220 s
Temperature at the midplane = 1399.93 °C

fx >>

ii) $K = 300 \text{ (W/m}\cdot\text{K)}$



Required time to cool the surface to 300°C = 207.260 s
Temperature at the midplane = 527.56 °C

fx >>

Upon examining the results, it was observed that the core temperature decreased only slightly relative to the initial temperature, while the surface cooled relatively rapidly due to convection (taking roughly 55 seconds to drop from 1400 °C to 300 °C). The reason the core did not cool sufficiently is that the thermal conductivity (K) was very low (a value of 30). For comparison purpose, when the K was increased to 300, the time required for the surface temperature to decrease to 300 °C was slightly longer - approximately 207 seconds - and the core temperature was calculated to be around 527 °C. Thus, a low thermal conductivity causes the thermal energy to remain within the slab rather than dissipating effectively.